

Assignment 1 Cloud Foundation and GCP Setup

CS 446 01

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1. ABSTRACT

This report documents the initial setup of a Google Cloud development environment and reviews core cloud computing concepts. The work included account access, project creation, budget alert configuration, and command line setup with the gcloud tool. Compute Engine API access was enabled to prepare for later virtual machine labs. The main thing is the early setup choices, especially billing controls and repeatable tooling, reduce risk and improve speed.

2. INTRODUCTION

Cloud computing provides computing resources such as compute, storage, and networking through on demand access. Instead of building and maintaining local infrastructure, teams can provision resources quickly and pay based on usage. This assignment focused on defining common cloud models, explaining cost differences in cloud adoption, and completing a first time Google Cloud setup. The work also established cost controls and a command line workflow that supports repeatable engineering tasks.

3. TASK WALKTHROUGH AND EXPLANATION

3.1. Service models. Infrastructure as a Service IaaS provides virtualized hardware resources such as virtual machines, disks, and networks. The customer manages the operating system, installed software, and application configuration. Google Compute Engine is an IaaS service because it provides virtual machines while leaving system management decisions to the user.

Platform as a Service PaaS provides a managed runtime environment for deploying applications. The provider manages the operating system layer and common platform components, while the user focuses on application code and configuration. Google App Engine fits PaaS because it runs deployed applications without requiring the user to manage servers directly.

Software as a Service SaaS delivers a complete application that users access through a browser or client. The provider manages the full stack and the user mainly configures the application and uses it. Google Docs fits SaaS because users consume the application without managing the underlying infrastructure.

3.2. Economic impact. Capital Expenditure CapEx is money spent up front to purchase long lived assets such as servers, storage systems, and networking equipment. Operational Expenditure OpEx is ongoing spending for services and operations such as subscriptions, maintenance, electricity,

and support. Cloud adoption usually shifts costs toward OpEx because resources are rented and billed over time instead of purchased and depreciated. Benefits include reduced up front investment, faster provisioning, and improved cost matching to actual demand. This shift also reduces the risk of over buying hardware that sits unused during low demand periods.

3.3. Deployment models. A hybrid deployment model is the best match for a financial services company with strict core application requirements and a new mobile application goal. The core banking system can remain in a private environment with stronger control over residency, access, and security policy. The customer mobile application can run in a public cloud environment where elastic scaling and managed services reduce delivery time. Hybrid design supports separation of risk and compliance needs while still enabling modern development and rapid scaling where it is appropriate.

4. RESULTS

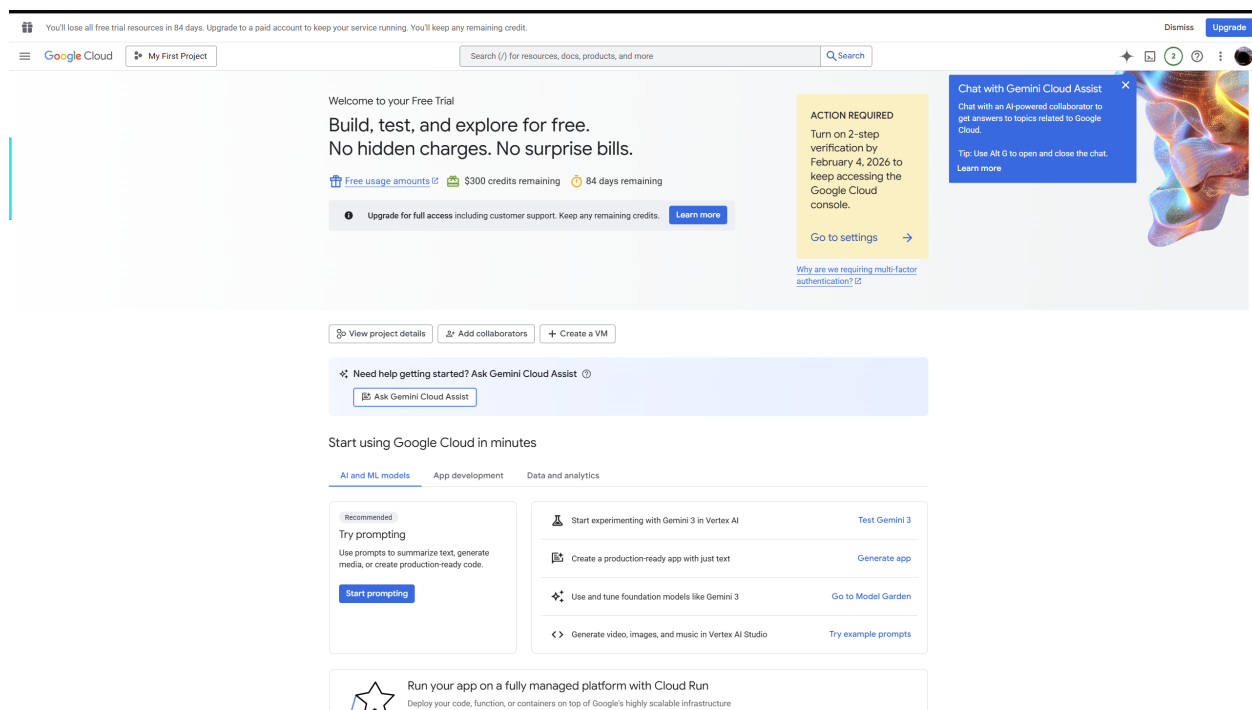


FIGURE 1. Google Cloud Console dashboard after initial login with free trial status visible

4.1. Account access.

4.2. Project creation.

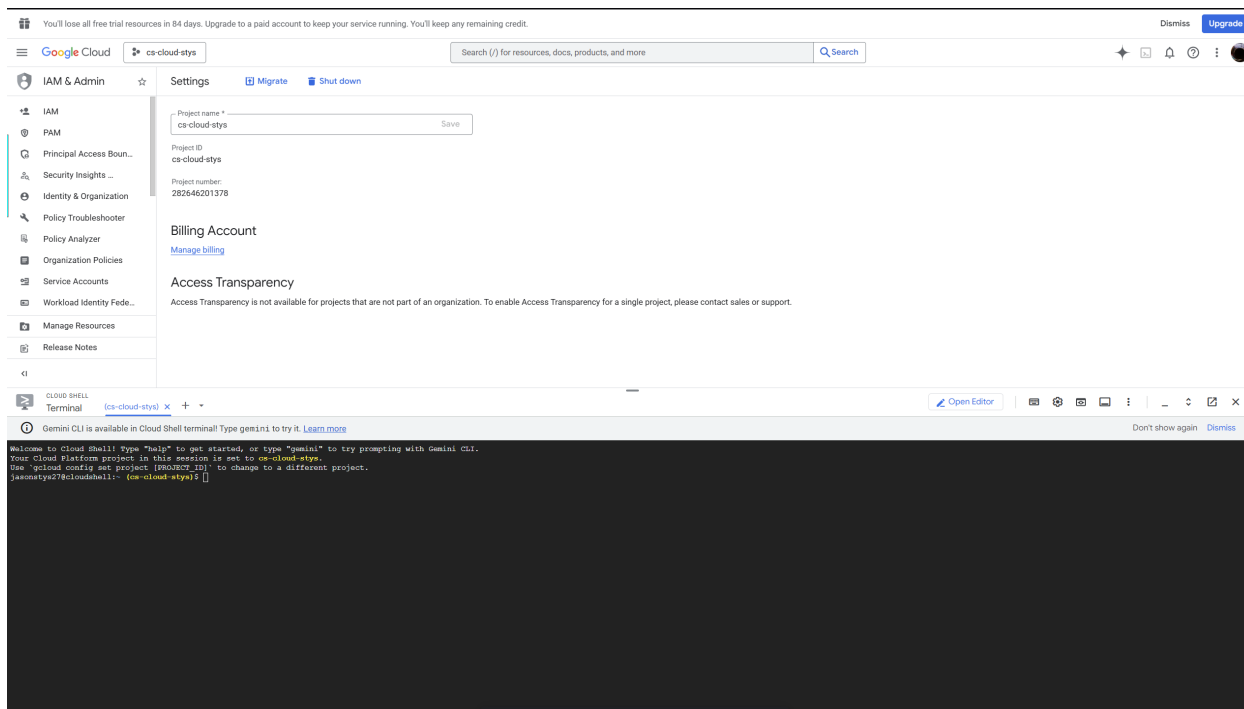


FIGURE 2. Project settings showing project name, project ID, and project number for cs cloud stys

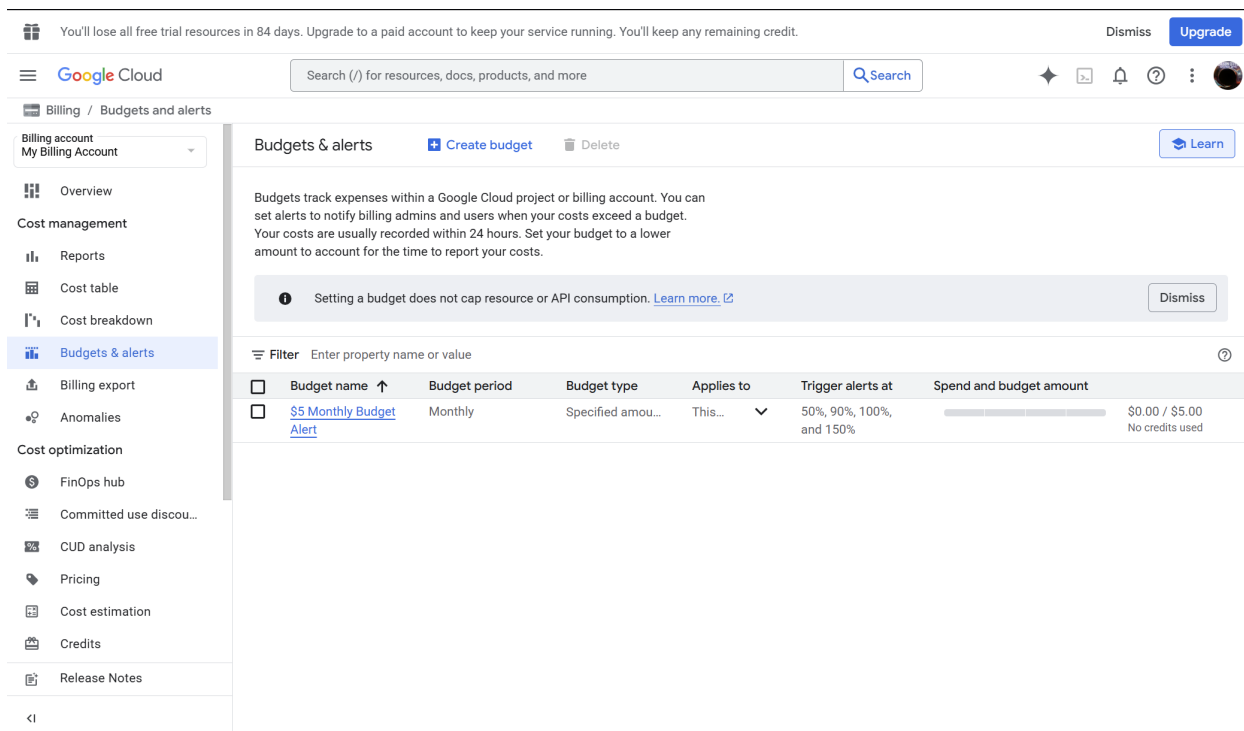


FIGURE 3. Billing budget list showing a five dollar monthly budget with alert thresholds configured

4.3. Budget alert setup.

```
Your current project has been set to: [cs-cloud-stys].

Not setting default zone/region (this feature makes it easier to use
[gcloud compute] by setting an appropriate default value for the
--zone and --region flag).
See https://cloud.google.com/compute/docs/gcloud-compute section on how to set
default compute region and zone manually. If you would like [gcloud init] to be
able to do this for you the next time you run it, make sure the
Compute Engine API is enabled for your project on the
https://console.developers.google.com/apis page.

Created a default .boto configuration file at [C:\Users\jason\boto]. See this file and
[https://cloud.google.com/storage/docs/gsutil/commands/config] for more
information about configuring Google Cloud Storage.
The Google Cloud CLI is configured and ready to use!

* Commands that require authentication will use jasonstys27@gmail.com by default
* Commands will reference project 'cs-cloud-stys' by default
Run 'gcloud help config' to learn how to change individual settings

This gcloud configuration is called [default]. You can create additional configurations if you work with multiple accounts and/or projects.
Run 'gcloud topic configurations' to learn more.

Some things to try next:

* Run 'gcloud --help' to see the Cloud Platform services you can interact with. And run 'gcloud help COMMAND' to get help on any gcloud command.
* Run 'gcloud topic --help' to learn about advanced features of the CLI like arg files and output formatting
* Run 'gcloud cheat-sheet' to see a roster of go-to 'gcloud' commands.

C:\Users\jason\AppData\Local\Google\Cloud SDK>gcloud config list
API [cloudresourcemanager.googleapis.com] not enabled on project [cs-cloud-stys]. Would you like to enable and retry
(this will take a few minutes)? (y/N)? y

Enabling service [cloudresourcemanager.googleapis.com] on project [cs-cloud-stys]...
Operation "operations/acet.p2-282646201378-f4cbe861-5edc-42cc-a42f-aa09fca7312c" finished successfully.
Project 'cs-cloud-stys' lacks an 'environment' tag. Please create or add a tag with key 'environment' and a value like 'Production', 'Development', 'Test', or 'Staging'. Add an 'e
nvironment' tag using 'gcloud resource-manager tags bindings create'. See https://cloud.google.com/resource-manager/docs/creating-managing-projects#designate_project_environments_
with_tags for details.
[accessibility]
screen_reader = False
[core]
account = jasonstys27@gmail.com
disable_usage_reporting = True
project = cs-cloud-stys

Your active configuration is: [default]

C:\Users\jason\AppData\Local\Google\Cloud SDK>
```

FIGURE 4. Terminal output for gcloud configuration showing the active account and selected project

4.4. gcloud CLI initialization.

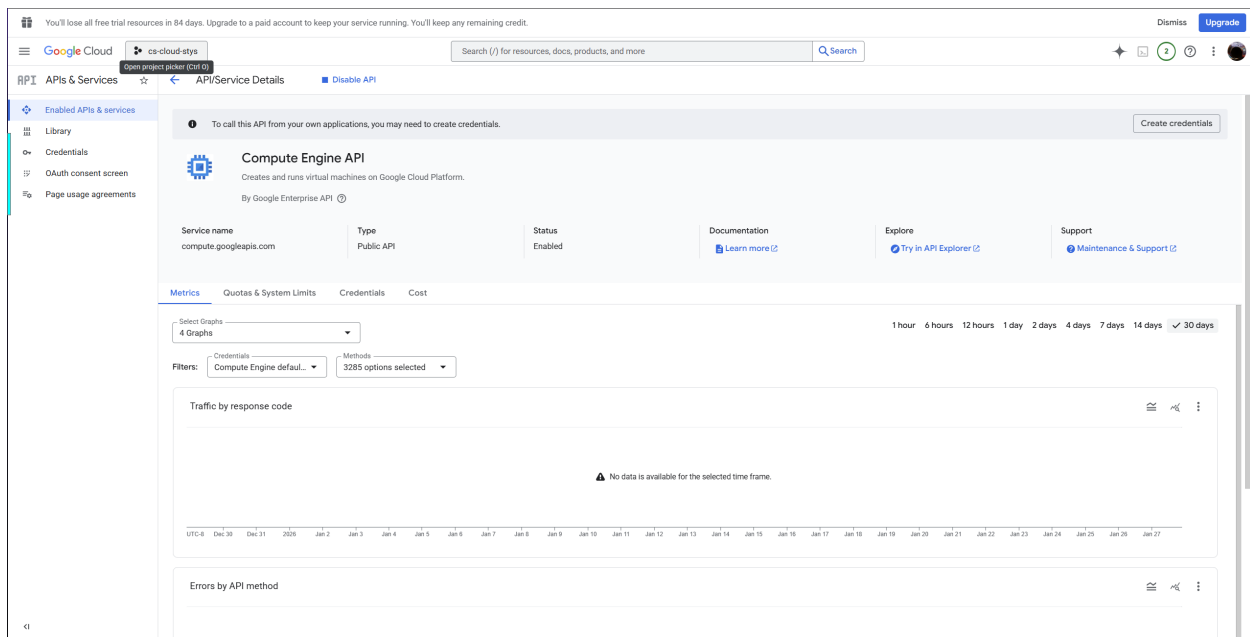
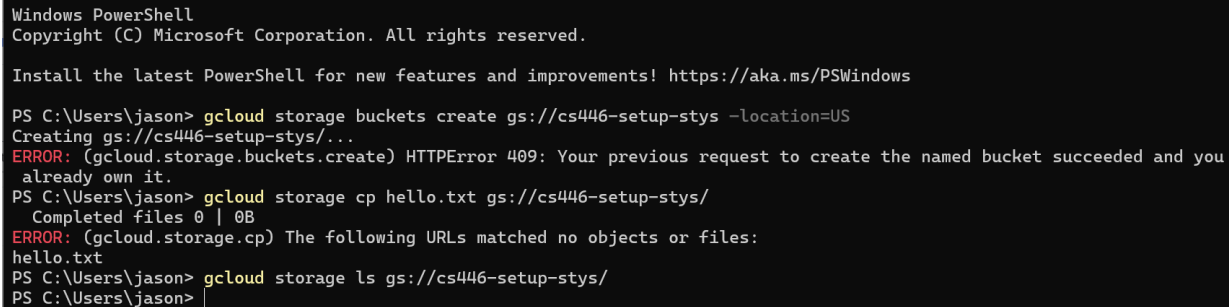


FIGURE 5. Compute Engine API service page showing enabled status

4.5. Compute Engine API enabled.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\jason> gcloud storage buckets create gs://cs446-setup-stys -location=US
Creating gs://cs446-setup-stys/...
ERROR: (gcloud.storage.buckets.create) HTTPError 409: Your previous request to create the named bucket succeeded and you already own it.
PS C:\Users\jason> gcloud storage cp hello.txt gs://cs446-setup-stys/
Completed files 0 | 0B
ERROR: (gcloud.storage.cp) The following URLs matched no objects or files:
hello.txt
PS C:\Users\jason> gcloud storage ls gs://cs446-setup-stys/
PS C:\Users\jason> |
```

FIGURE 6. PowerShell session running gcloud storage commands for bucket creation and file upload workflow

4.6. Cloud Storage CLI commands.

5. DISCUSSION

The most confusing part of the setup was the number of places where configuration can happen. Console settings, Cloud Shell defaults, and local Cloud SDK settings all affect the active project and enabled services. And when the command line prompted to enable an API, it highlighted that permissions and API state can block progress even when authentication is correct. I am not sure if this was the reason why I was unable to get part 3 fully working or not, but its my only guess. Despite the fact that I had the hello.txt file the command to put the file in the cloud storage didn't seem to work, with it being unable to find a connection to the file on my PC, likely do to not having access, why it cant I do not fully know.

Budget alerts matter because they provide early warning before costs grow beyond what was intended. A small budget limit reduces risk during learning and prevents surprise bills from misconfigured resources. This is important for individuals and for companies because it adds accountability and supports consistent cost control practices. (2)

The gcloud command line is faster for repeatable tasks and it supports scripting, automation, and version controlled workflows. The console is often better for discovery and for reading details across many services. A mixed approach works well, using the console to learn and verify, then use the command line to execute repeatable work. (4) (3) (1)

6. CONCLUSION

A working Google Cloud project with basic governance and tooling in place. The setup included project creation, budget alerts, and a verified gcloud configuration tied to the correct account and project. Compute Engine API was enabled to support future virtual machine labs. The conceptual questions connected cloud service models and deployment choices to real constraints such as cost control and compliance.

7. REFERENCES

REFERENCES

1. Google Cloud, *gcloud storage command group* (2026; <https://cloud.google.com/sdk/gcloud/reference/storage>).
2. Google Cloud, *Google Cloud Billing budgets and alerts* (2026; <https://cloud.google.com/billing/docs/how-to/budgets>).
3. Google Cloud, *Initialize the gcloud CLI* (2026; <https://cloud.google.com/sdk/gcloud/reference/init>).
4. Google Cloud, *Install the gcloud CLI* (2026; <https://cloud.google.com/sdk/docs/install>).

8. APPENDIX

8.1. Command line summary.

```
gcloud init
```

```
gcloud config list
```

```
gcloud storage buckets create gs://cs446-setup-stys --location=US
```

```
gcloud storage cp hello.txt gs://cs446-setup-stys/
```

```
gcloud storage ls gs://cs446-setup-stys/
```
